

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Database Management Systems
COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO3: Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)

Activity Tool: MCQ Test (Low)
Assessment Tool Weightage: 30%

Code of Conduct:

I, A Mohammad Naheed-192525336, certify that this submission is my original work and that I have followed the guidelines specified for this assessment.

I understand that violation of academic integrity rules may result in disciplinary action.

Signature of the Student: A Mohammad Naheed

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key reference	BL2 – Understand	5

8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	5
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	5
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	5

Q1. What is the main requirement of First Normal Form (1NF)?

- (a) Each table must have a foreign key
- (b) Every attribute should contain a single atomic value
- (c) All attributes must be candidate keys
- (d) Transitive dependencies must be removed

✓ **Correct Answer: (b) Every attribute should contain a single atomic value**

Explanation:

A relation satisfies 1NF when each field contains only one indivisible value. Repeating groups and multiple values stored in a single column should be avoided. Partial dependencies are handled by 2NF, while transitive dependencies are addressed by 3NF.

SQL Demonstration:

```
CREATE TABLE StudentContact (
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(30),
    Phone VARCHAR(15)
);
```

```
INSERT INTO StudentContact VALUES
(101, 'Rahul', '9876543210');
```

```
+-----+-----+-----+
| StudentID | StudentName | Phone |
+-----+-----+-----+
| 101 | Rahul | 9876543210 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

Q2. Which condition must be satisfied for a relation to be in 2NF?

- (a) It must have no multivalued dependencies
- (b) It must have no partial dependency on a composite key
- (c) Every attribute must be a prime attribute
- (d) It must contain only one candidate key

✓ **Correct Answer: (b) It must have no partial dependency on a composite key**

Explanation:

2NF requires a relation to already satisfy 1NF and ensures that every non-key attribute depends on the complete primary key. Partial dependency occurs when a non-key attribute depends on only a portion of a composite key.

SQL Demonstration:

```
CREATE TABLE Enrollment (  
    StudentID INT,  
    CourseID INT,  
    Grade CHAR(2),  
    PRIMARY KEY (StudentID, CourseID)  
);
```

Field	Type	Null	Key	Default	Extra
StudentID	int	NO	PRI	NULL	
CourseID	int	NO	PRI	NULL	
Grade	char(2)	YES		NULL	

3 rows in set (0.04 sec)

Q3. Which normal form is specifically designed to eliminate multivalued dependencies?

- (a) 2NF
- (b) 3NF
- (c) BCNF
- (d) 4NF

✓ **Correct Answer: (d) 4NF**

Explanation:

Fourth Normal Form (4NF) deals with multivalued dependencies. It helps remove redundancy when two or more independent sets of values are associated with the same entity.

SQL Demonstration:

```
CREATE TABLE EmployeeSkill (  
    EmpID INT,  
    Skill VARCHAR(30)  
);
```

```
CREATE TABLE EmployeeLanguage (  
    EmpID INT,  
    Language VARCHAR(30)  
);
```

```
+-----+-----+-----+-----+-----+-----+  
| Field | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| EmpID | int           | YES  |     | NULL    |      |  
| Skill | varchar(30)   | YES  |     | NULL    |      |  
+-----+-----+-----+-----+-----+-----+  
2 rows in set (0.00 sec)
```

```
mysql> desc EmployeeLanguage;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| EmpID | int           | YES  |     | NULL    |      |  
| Language | varchar(30) | YES  |     | NULL    |      |  
+-----+-----+-----+-----+-----+-----+  
2 rows in set (0.00 sec)
```

Q4. Boyce-Codd Normal Form (BCNF) can be considered a stronger form of which normal form?

- (a) 1NF
- (b) 2NF
- (c) 3NF
- (d) 5NF

✓ **Correct Answer: (c) 3NF**

Explanation:

BCNF is stricter than 3NF. In BCNF, every determinant must be a candidate key. Therefore, a relation satisfying BCNF will also satisfy 3NF, although a relation in 3NF may not necessarily satisfy BCNF.

SQL Demonstration:

```
CREATE TABLE FacultyCourse (  
    FacultyID INT,  
    CourseID INT,  
    FacultyName VARCHAR(30)  
);
```

Field	Type	Null	Key	Default	Extra
FacultyID	int	YES		NULL	
CourseID	int	YES		NULL	
FacultyName	varchar(30)	YES		NULL	

3 rows in set (0.00 sec)

Q5. Consider the functional dependency $P \rightarrow Q$. What does this notation indicate?

- (a) Q uniquely determines P
- (b) P uniquely determines Q
- (c) P and Q have no relationship
- (d) P and Q always determine each other

✓ **Correct Answer: (b) P uniquely determines Q**

Explanation:

The functional dependency $P \rightarrow Q$ means that whenever two tuples have the same value of P, they must also have the same value of Q. In other words, P determines Q.

SQL Demonstration:

```
CREATE TABLE Student (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(30)  
);
```

Field	Type	Null	Key	Default	Extra
StudentID	int	NO	PRI	NULL	
StudentName	varchar(30)	YES		NULL	

2 rows in set (0.00 sec)

Q6. Which of the following is NOT a commonly recognized database anomaly caused by redundancy?

- (a) Insertion anomaly
- (b) Modification anomaly
- (c) Deletion anomaly
- (d) Retrieval anomaly

✓ Correct Answer: (d) Retrieval anomaly

Explanation:

The three commonly discussed anomalies caused by redundant data are insertion, update/modification, and deletion anomalies. Normalization aims to reduce these problems by organizing data into appropriate relations.

SQL Demonstration:

Example: If the same student's department information is stored repeatedly in many records, changing the department name in only some rows can produce inconsistent data. This is an update anomaly.

Q7. Which statement correctly defines a superkey?

- (a) A foreign key that references another table
- (b) The smallest possible unique identifier
- (c) Any attribute set that can uniquely identify a tuple
- (d) An attribute that has a transitive dependency

✓ Correct Answer: (c) Any attribute set that can uniquely identify a tuple

Explanation:

A superkey is a set of one or more attributes that uniquely identifies each tuple in a relation. It does not have to be minimal. For example, {StudentID} is a superkey, and {StudentID, StudentName} can also be a superkey although it is not minimal.

Q8. Given the dependencies $X \rightarrow Y$ and $Y \rightarrow Z$, what type of dependency exists between X and Z?

- (a) Partial dependency
- (b) Multivalued dependency
- (c) Transitive dependency
- (d) Independent dependency

✓ **Correct Answer: (c) Transitive dependency**

Explanation:

Since X determines Y and Y determines Z, X can indirectly determine Z. The dependency of Z on X through Y is called a transitive dependency. 3NF aims to eliminate such dependencies.

SQL Demonstration:

```
CREATE TABLE StudentDepartment (  
    StudentID INT PRIMARY KEY,  
    DepartmentID INT,  
    DepartmentName VARCHAR(40)  
) ;
```

Field	Type	Null	Key	Default	Extra
StudentID	int	NO	PRI	NULL	
DepartmentID	int	YES		NULL	
DepartmentName	varchar(40)	YES		NULL	

3 rows in set (0.01 sec)

Q9. What is another name for Fifth Normal Form (5NF)?

- (a) Domain-Key Normal Form
- (b) Boyce-Codd Normal Form
- (c) Project-Join Normal Form
- (d) Atomic Normal Form

✓ **Correct Answer: (c) Project-Join Normal Form**

Explanation:

5NF is also called Project-Join Normal Form (PJNF). It deals mainly with join dependencies and ensures that a relation cannot be further decomposed without introducing loss of information or requiring a join dependency not implied by candidate keys.

SQL Demonstration:

Example: A Supplier-Part-Project relationship may require decomposition into smaller relations while preserving the original information.

Q10. What is the primary objective of a lossless-join decomposition?

- (a) To ensure that no information is lost after decomposition
- (b) To eliminate all foreign keys
- (c) To guarantee that every relation is in BCNF
- (d) To remove all NULL values

✓ **Correct Answer: (a) To ensure that no information is lost after decomposition**

Explanation:

A decomposition is lossless when joining the decomposed relations produces exactly the original relation. It prevents the generation of incorrect or spurious tuples and ensures that original information can be recovered.

SQL Demonstration:

```
CREATE TABLE StudentInfo (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(30)  
);
```

```
CREATE TABLE StudentDepartmentInfo (  
    StudentID INT,  
    DepartmentID INT  
);
```

Field	Type	Null	Key	Default	Extra
StudentID	int	NO	PRI	NULL	
StudentName	varchar(30)	YES		NULL	

2 rows in set (0.00 sec)

```
mysql> desc StudentDepartmentInfo;
```

Field	Type	Null	Key	Default	Extra
StudentID	int	YES		NULL	
DepartmentID	int	YES		NULL	

2 rows in set (0.00 sec)